

# MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics  
California State University Fullerton

August 23, 2026

# Outline

- 1 SoPS Lecture 1
  - Welcome
  - New game
  - New game plus
  - Finishing up

# Outline

- 1 SoPS Lecture 1
  - Welcome
  - New game
  - New game plus
  - Finishing up

# What is this course about?

# What is this course about?

Learn what mathematicians do

# What is this course about?

Learn what mathematicians do

- play → discover patterns

# What is this course about?

## Learn what mathematicians do

- play  $\rightarrow$  discover patterns
- abstract  $\rightarrow$  express patterns mathematically

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true



# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

- logic, set theory, cardinality

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

## Make friends

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

## Make friends

- math is **social**

# What is this course about?

## Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

## Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

## Make friends

- math is **social**
- you'll see each other *a lot*

# Course expectations

# Course expectations

## Grading



# Course expectations

## Grading

- pre-class assignments  $\rightarrow$  15%

# Course expectations

## Grading

- pre-class assignments → 15%
- in-class assignments → 25%

# Course expectations

## Grading

- pre-class assignments → 15%
- in-class assignments → 25%
- homework → 25%

# Course expectations

## Grading

- pre-class assignments → 15%
- in-class assignments → 25%
- homework → 25%
- seminars / research → 15%

# Course expectations

## Grading

- pre-class assignments → 15%
- in-class assignments → 25%
- homework → 25%
- seminars / research → 15%
- final exam → 20%

# Course expectations

# Course expectations

## Daily Expectations

# Course expectations

## Daily Expectations

- complete pre-class assignments before class



# Course expectations

## Daily Expectations

- complete pre-class assignments before class
- attend every class

# Course expectations

## Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**

# Course expectations

## Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**
- solicit participation from group members

# Course expectations

## Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**
- solicit participation from group members
- do your homework (or else)

# Course expectations

# Course expectations

## Seminar Participation

# Course expectations

## Seminar Participation

- must attend at least three

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report



# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

- pure math seminar

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

- pure math seminar
- applied math seminar

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar
- problem solving seminar

# Course expectations

## Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

## Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar
- problem solving seminar
- other events (workshops, colloquia, etc)

# Course expectations



# Course expectations

## Research project

# Course expectations

## Research project

- what math research is

# Course expectations

## Research project

- what math research is
- do original research

# Course expectations

## Research project

- what math research is
- do original research
- write a final report

# Course expectations

## Research project

- what math research is
- do original research
- write a final report
- present results

# Outline

- 1 SoPS Lecture 1
  - Welcome
  - **New game**
  - New game plus
  - Finishing up

# Pinching Pennies

# Pinching Pennies

## Rules



# Pinching Pennies

## Rules

- arrange pennies in two piles

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

## Activity (20 mins)

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

## Activity (20 mins)

- split into pairs

# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

## Activity (20 mins)

- split into pairs
- play 4-5 rounds



# Pinching Pennies

## Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

## Activity (20 mins)

- split into pairs
- play 4-5 rounds
- **record data:** initial board, who won?

# Pinching Pennies: Debrief

# Pinching Pennies: Debrief

**Let's combine the data ...**

# Pinching Pennies: Debrief

**Let's combine the data ...**

**Question 1**

is it better to be **Player 1** or **Player 2** ?

# Pinching Pennies: Debrief

**Let's combine the data ...**

## Question 1

is it better to be **Player 1** or **Player 2** ?

## Question 2

if the piles are equal, then who should win?

# Pinching Pennies: Debrief

**Let's combine the data ...**

**Question 1**

is it better to be **Player 1** or **Player 2** ?

**Question 2**

if the piles are equal, then who should win?

**Fight me!**

# Vocabulary

# Vocabulary

## Optimal Play



# Vocabulary

## Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

# Vocabulary

## Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

## Winning/Losing Position

# Vocabulary

## Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

## Winning/Losing Position

A **winning position** for a player in a game is a state of the game where if that player plays optimally, they are guaranteed to win. A **losing position** for a player is a game state where one of their opponents has a winning position.

# Outline

- 1 SoPS Lecture 1
  - Welcome
  - New game
  - **New game plus**
  - Finishing up

# Pinching Pennies – revisited

# Pinching Pennies – revisited

Activity (20 minutes)

# Pinching Pennies – revisited

## Activity (20 minutes)

- arrange pennies in **three piles**

# Pinching Pennies – revisited

## Activity (20 minutes)

- arrange pennies in **three piles**
- which starts are losing positions for **Player 1**?



# Pinching Pennies – revisited

## Activity (20 minutes)

- arrange pennies in **three piles**
- which starts are losing positions for **Player 1**?

## Submit your conjecture:



# Break time!

## Activity (5 minutes)

- get up, stretch get a cup of coffee
- chat up your neighbor, introduce yourself

# Pinching Pennies – two equal piles

# Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

# Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

# Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

Question 3:

Two piles with three pennies each, who wins?

# Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

Question 3:

Two piles with three pennies each, who wins?

Copypat strategy

**Player 2** copies what **Player 1** does to the opposite pile

# Our first proof



# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

If **Player 1** takes a whole pile, **Player 2** wins.

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

If **Player 1** takes a whole pile, **Player 2** wins.

Otherwise **Player 1** must reduce a pile to  $m$  pennies with  $m < n$ .

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

If **Player 1** takes a whole pile, **Player 2** wins.

Otherwise **Player 1** must reduce a pile to  $m$  pennies with  $m < n$ .

Then **Player 2** reduces the other pile to  $m$  pennies.

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

If **Player 1** takes a whole pile, **Player 2** wins.

Otherwise **Player 1** must reduce a pile to  $m$  pennies with  $m < n$ .

Then **Player 2** reduces the other pile to  $m$  pennies.

Now we're back where we started, but with less pennies!

# Our first proof

## Theorem

Two equal piles of pennies is a losing position for **Player 1**.

## Proof

Start with  $n$  pennies in each pile.

If **Player 1** takes a whole pile, **Player 2** wins.

Otherwise **Player 1** must reduce a pile to  $m$  pennies with  $m < n$ .

Then **Player 2** reduces the other pile to  $m$  pennies.

Now we're back where we started, but with less pennies!

Repeat until **Player 1** is forced to take a whole pile.



# Outline

- 1 SoPS Lecture 1
  - Welcome
  - New game
  - New game plus
  - Finishing up

# Question of the Day

## Puzzle

You have 15 pennies. On each turn, a player removes 1, 2, or 3 coins. The player who takes the last coin wins. Is the start a winning position for **Player 1** or **Player 2**?

# Final thoughts

## Reflection

- What did you notice about how mathematicians work compared to how you expected?
- What differences do you see between experimentation (play) and proof?
- How is this class different from the math you've done before? What do you think is the key to your success in this class?

Submit your responses by **tonight!**